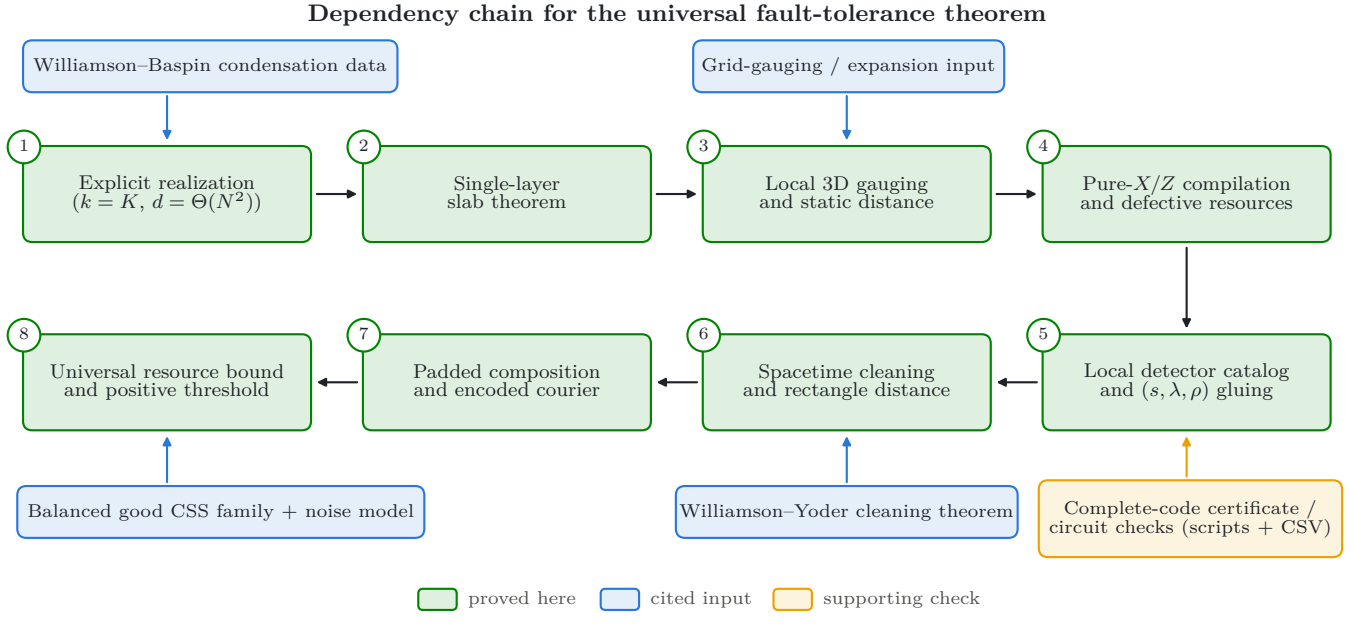

Dependency chain for the universal fault-tolerance theorem
Williamson–Baspin condensation data
Grid-gauging / expansion input
1
Explicit realization
($k = K$, $d = \Theta(N^2)$)
2
Single-layer
slab theorem
3
Local 3D gauging
and static distance
4
Pure-X/Z compilation
and defective resources
5
Local detector catalog
and (s, λ, ρ) gluing
6
Spacetime cleaning
and rectangle distance
7
Padded composition
and encoded courier
8
Universal resource bound
and positive threshold
Balanced good CSS family + noise model
Williamson–Yoder cleaning theorem
Complete-code certificate /
circuit checks (scripts + CSV)
proved here
cited input
supporting check
Finite numerical crossings test selected models; they are not inputs to the theorem.